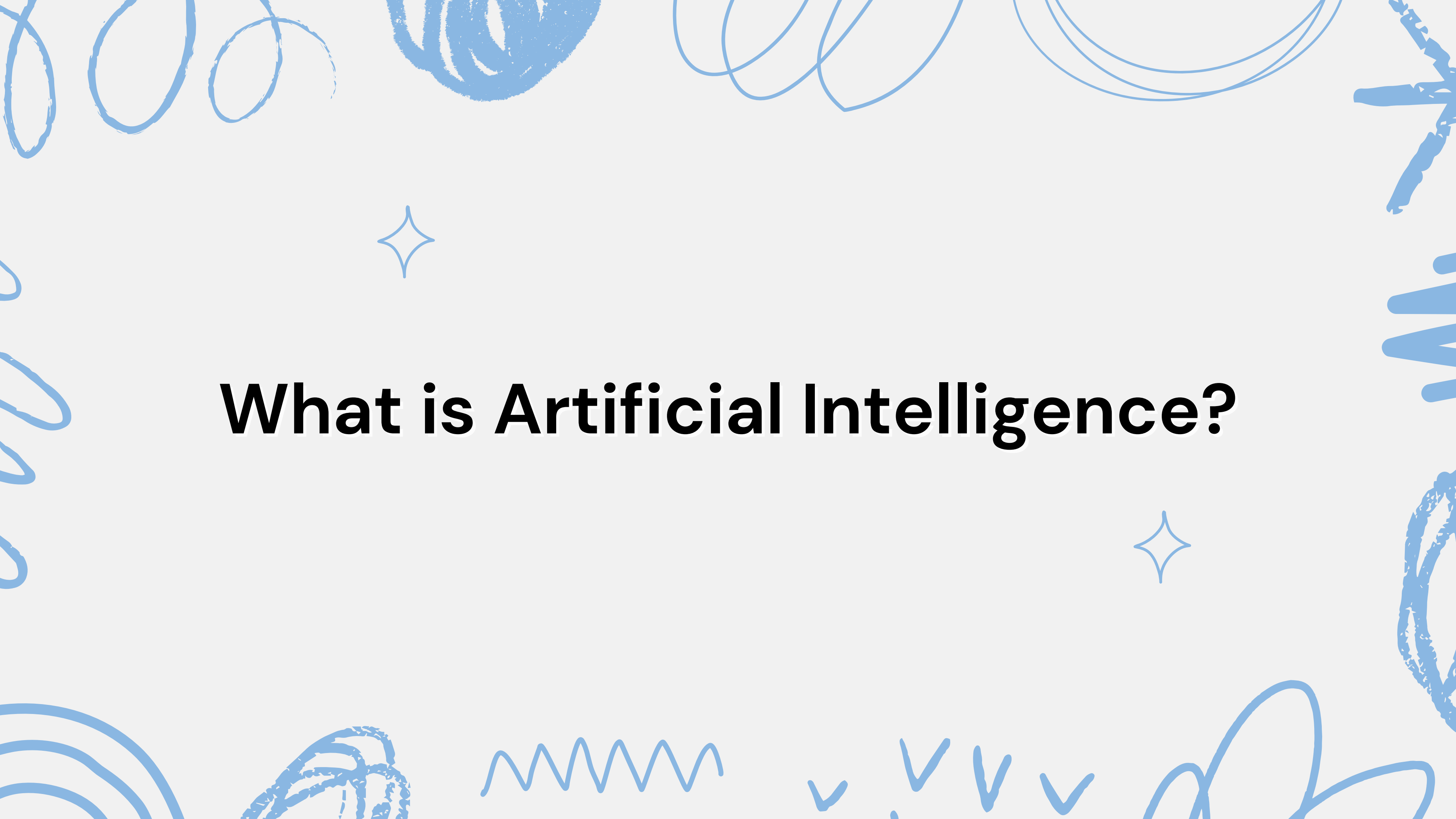




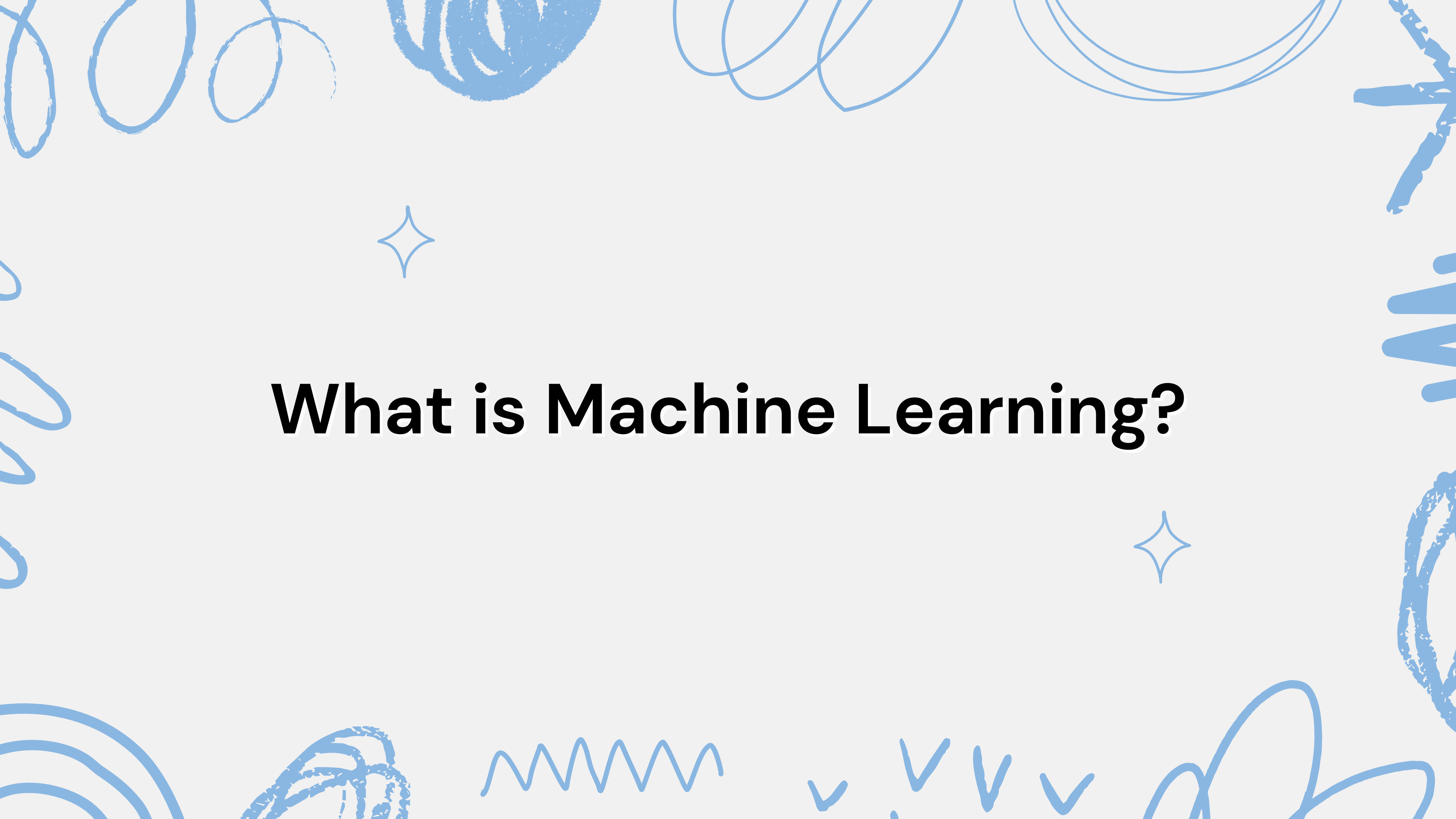
ML Intro, Feature Engineering & Accuracy Metrics

MSP'26 ML Module Session01

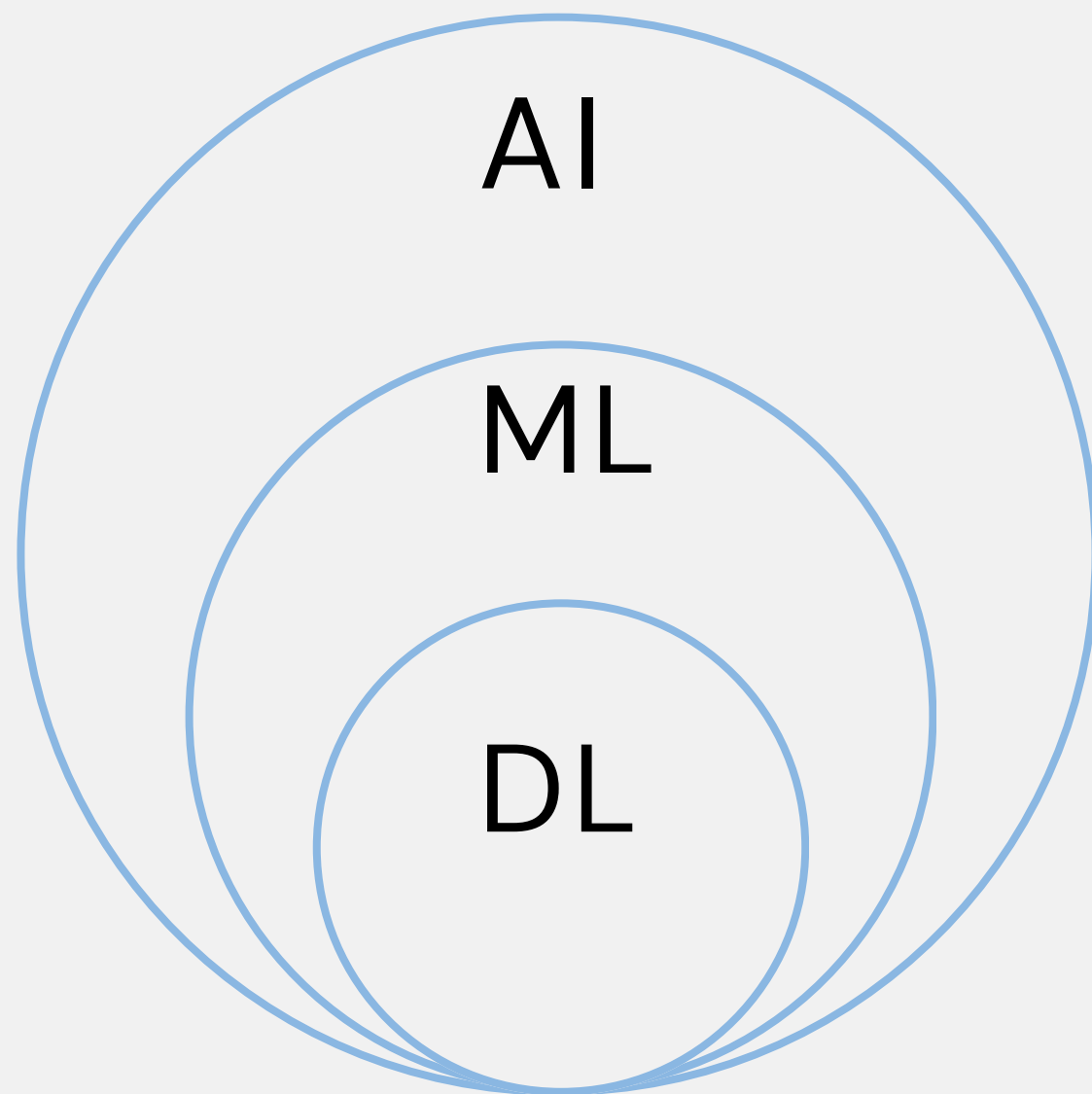




What is Artificial Intelligence?



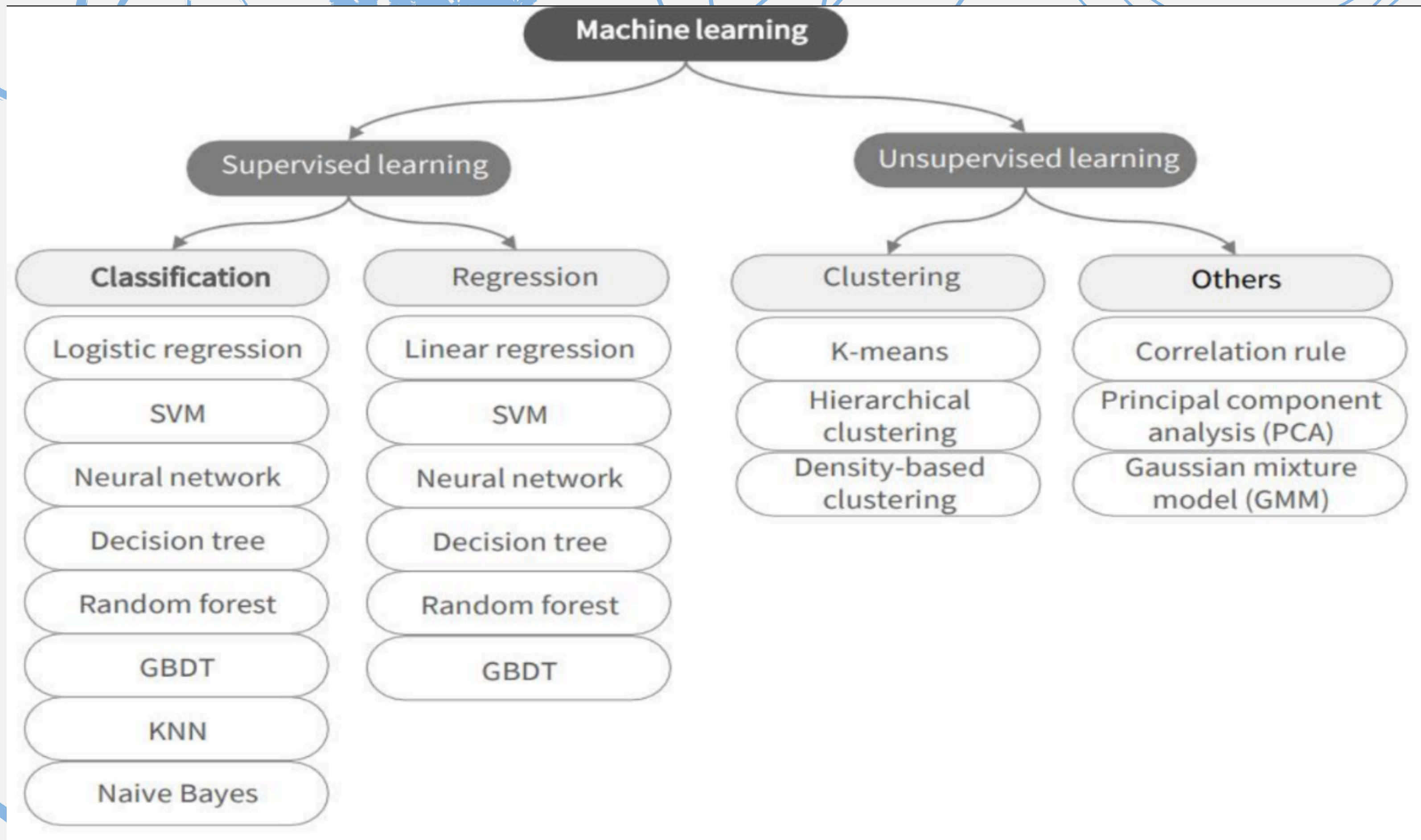
What is Machine Learning?



AI

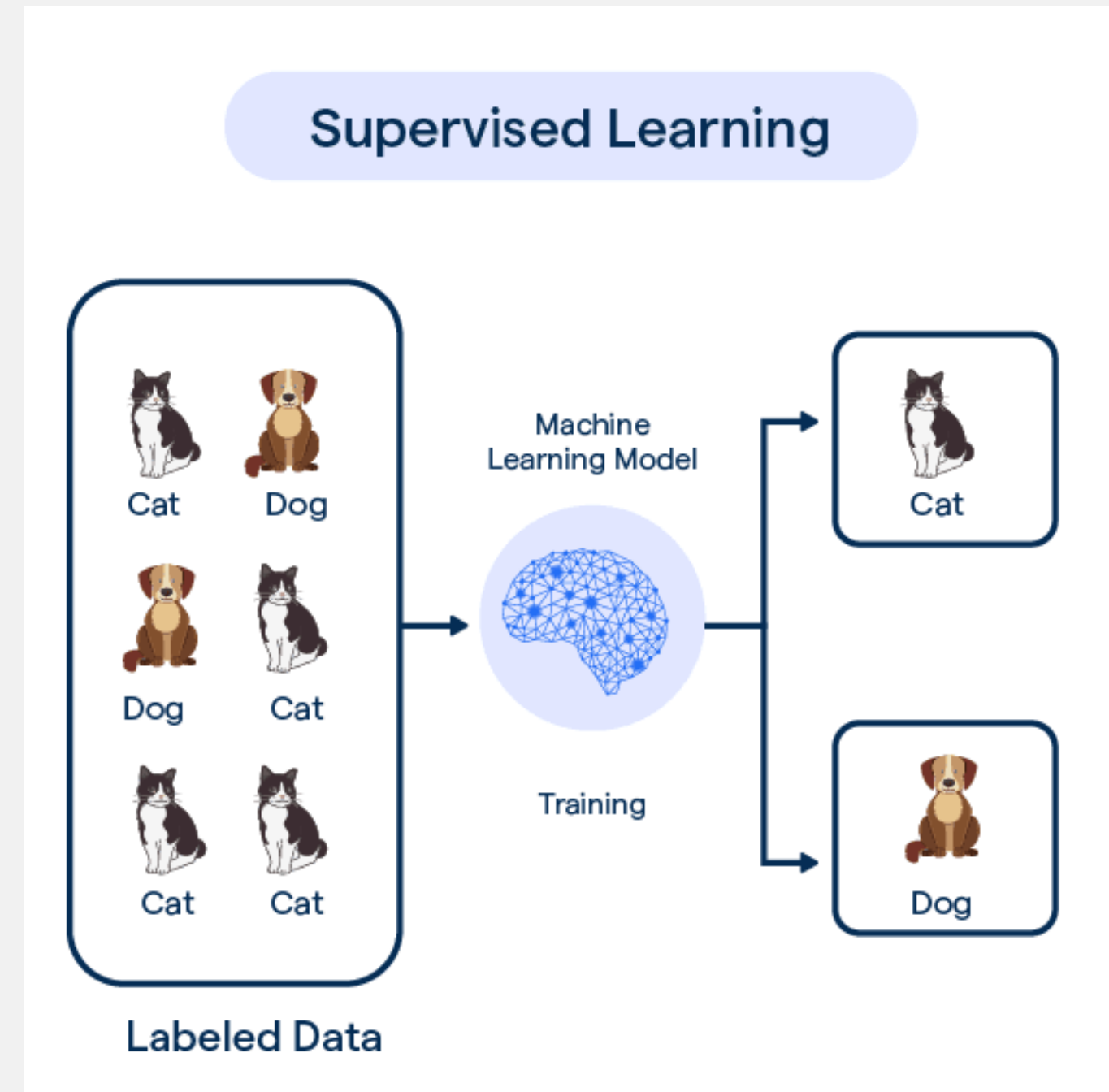
ML

DL



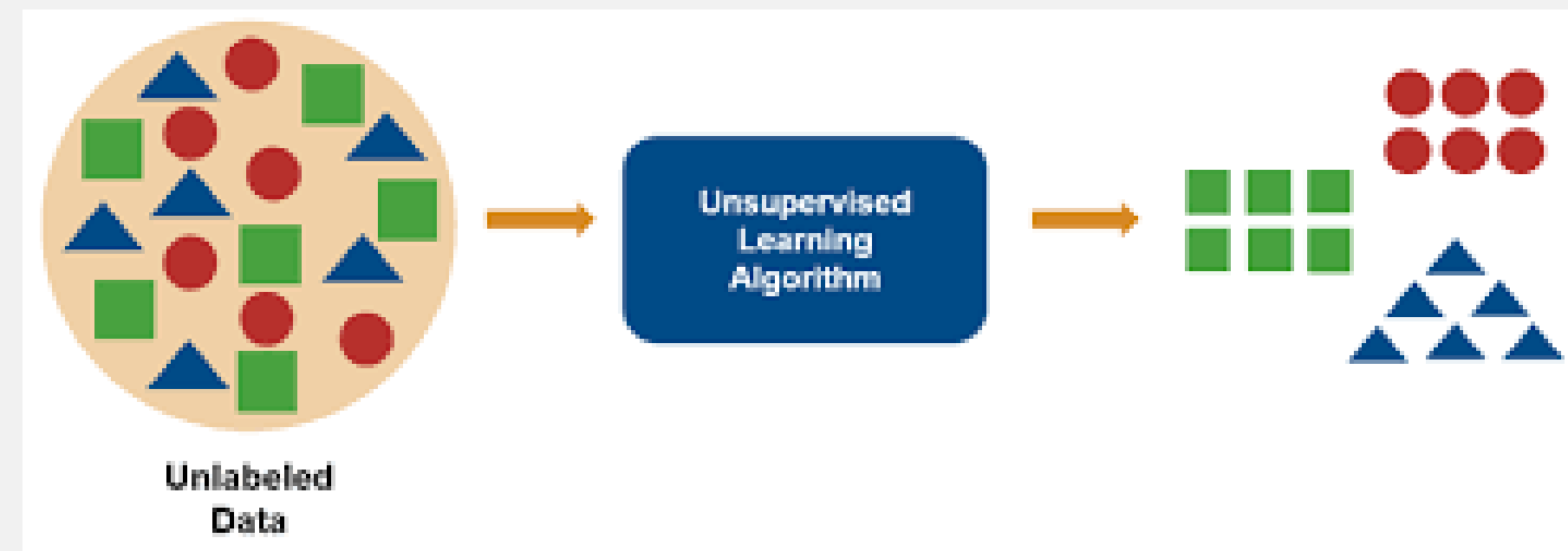
Supervised Learning

Supervised learning is a type of machine learning where a model learns from labelled data—meaning every input has a corresponding correct output. The model makes predictions and compares them with the true outputs, adjusting itself to reduce errors and improve accuracy over time. The goal is to make accurate predictions on new, unseen data.



Unsupervised Learning

Unsupervised Learning is a type of machine learning where the model works without labelled data. It learns patterns on its own by grouping similar data points or finding hidden structures without any human intervention.

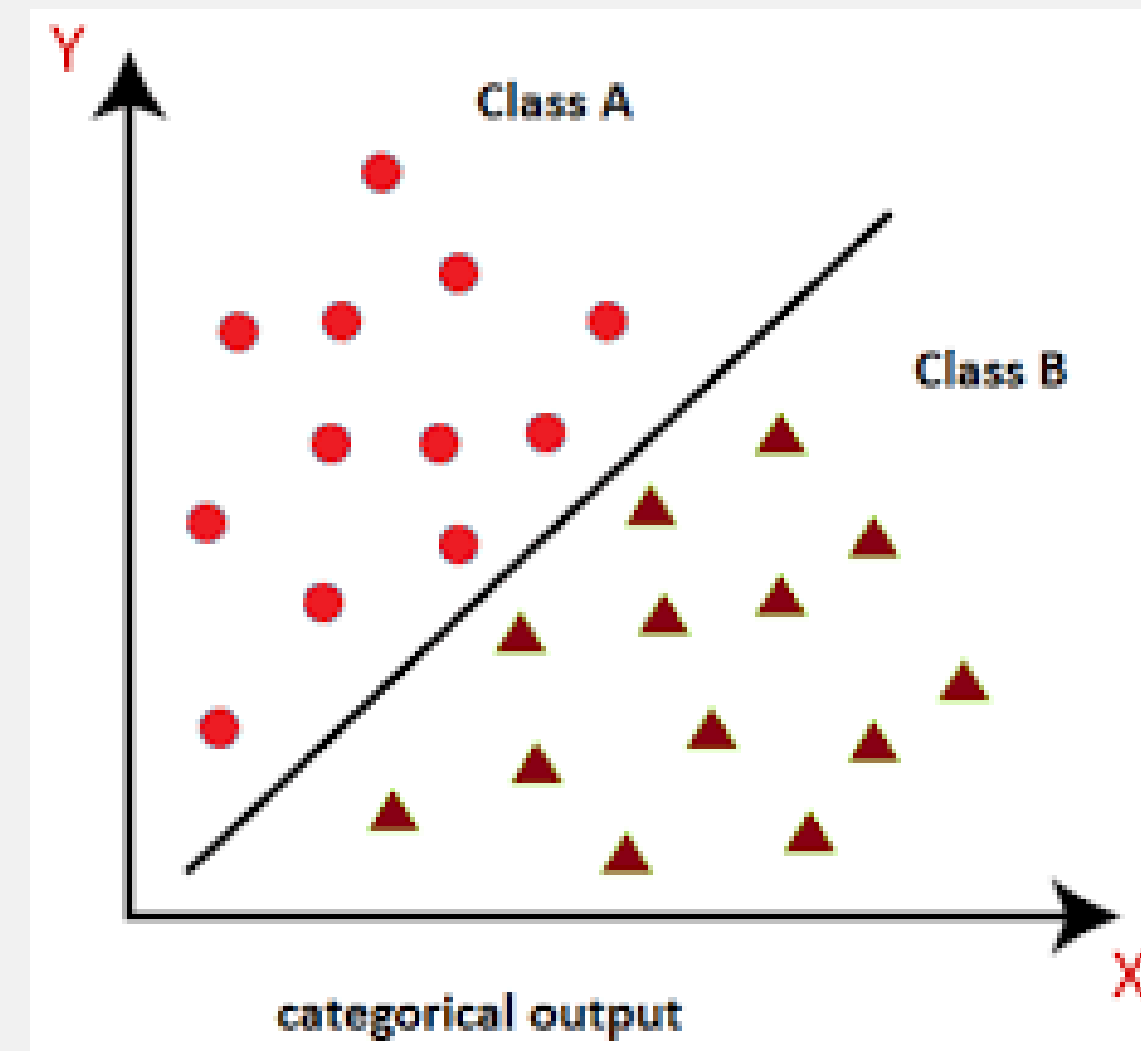




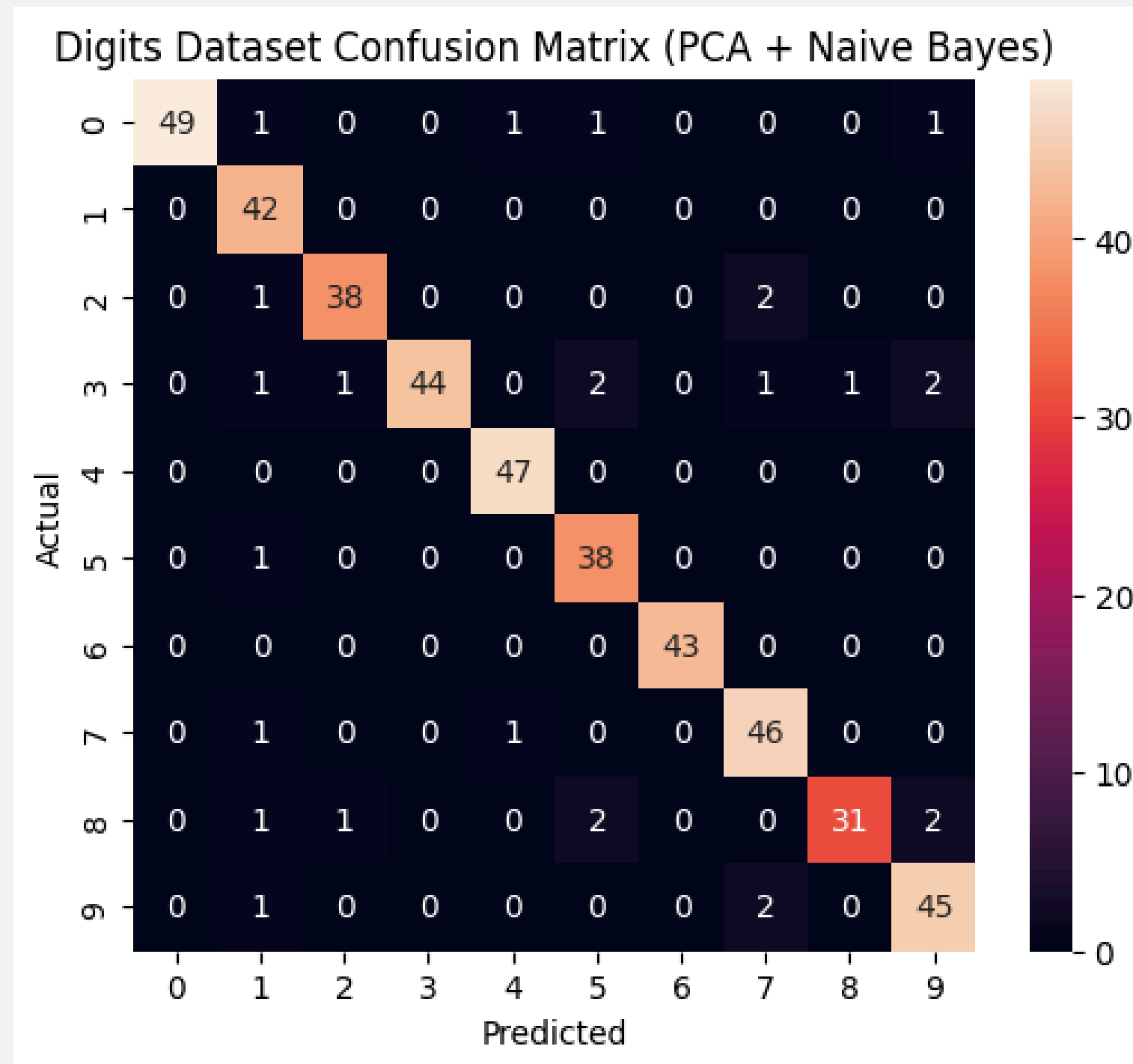
Supervised Learning

Supervised Learning – Classification

Classification is a supervised machine learning technique used to predict labels or categories based on input data. The goal is to assign each data point to a predefined class, such as spam vs. non-spam emails or diseased vs. healthy patients.



Classification Accuracy Metrics

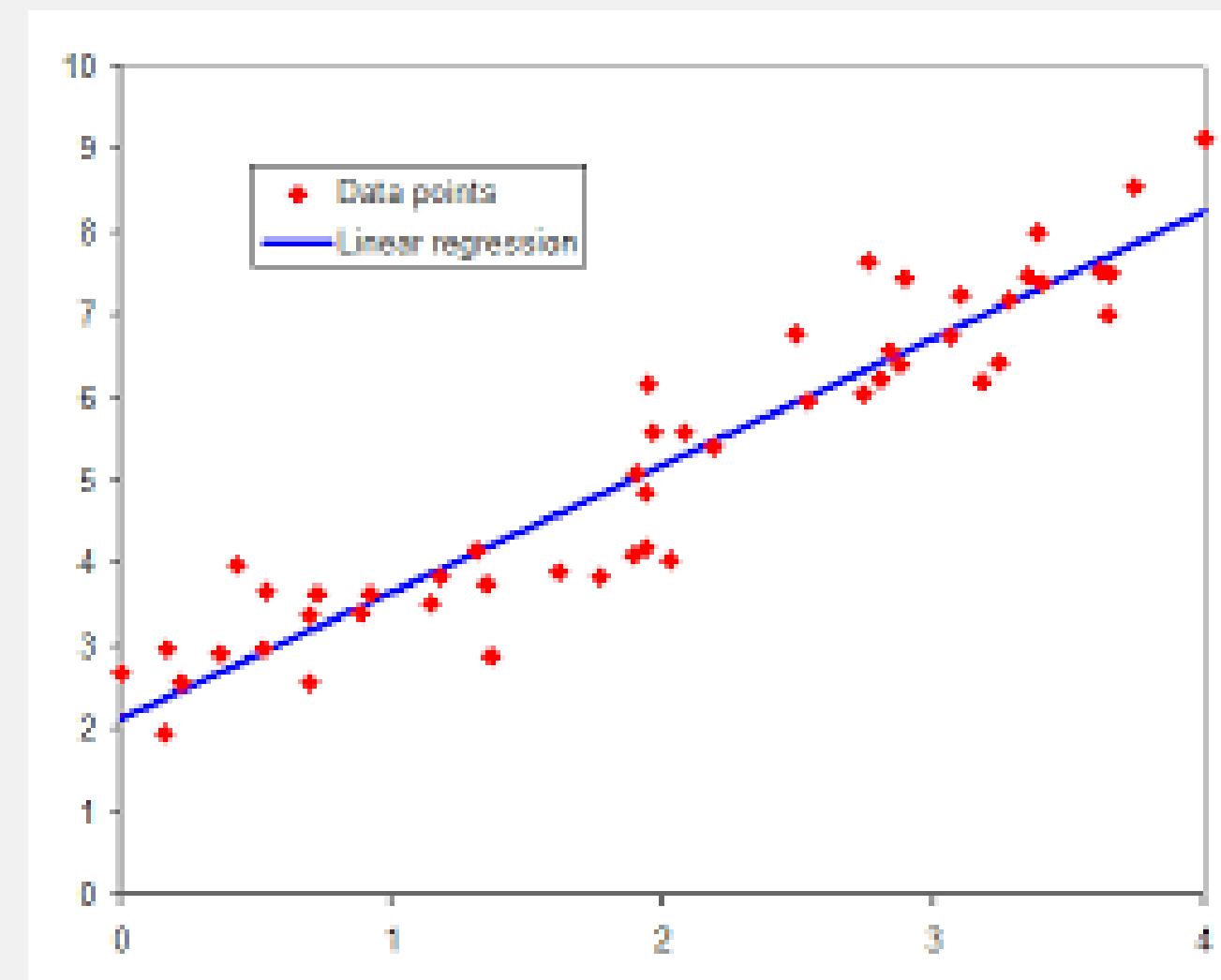


Classification Accuracy Metrics

Measurement	Formula
Accuracy, recognition rate	$\frac{TP + TN}{P + N}$
Error rate, misclassification rate	$\frac{FP + FN}{P + N}$
True positive rate, sensitivity, recall	$\frac{TP}{P}$
True negative rate, specificity	$\frac{TN}{N}$
Precision	$\frac{TP}{TP + FP}$
F_1 value, harmonic mean of precision and recall	$\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

Supervised Learning – Regression

Regression in machine learning is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target). It helps understand how changes in one or more factors influence a measurable outcome and is widely used in forecasting, risk analysis, decision-making and trend estimation.



Regression Accuracy Metrics

Mean Absolute Error (MAE)

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$



Unsupervised Learning

Clustering Accuracy Metrics

Silhouette Score

Measures how similar a point is to its own cluster vs others

Range: $-1 \rightarrow 1$

close to 1 $\rightarrow$ good clustering

close to 0 $\rightarrow$ overlapping clusters

negative $\rightarrow$ wrong clustering

Davies-Bouldin Index (DBI)

Measures cluster similarity

Lower is better

Penalizes clusters that are close together

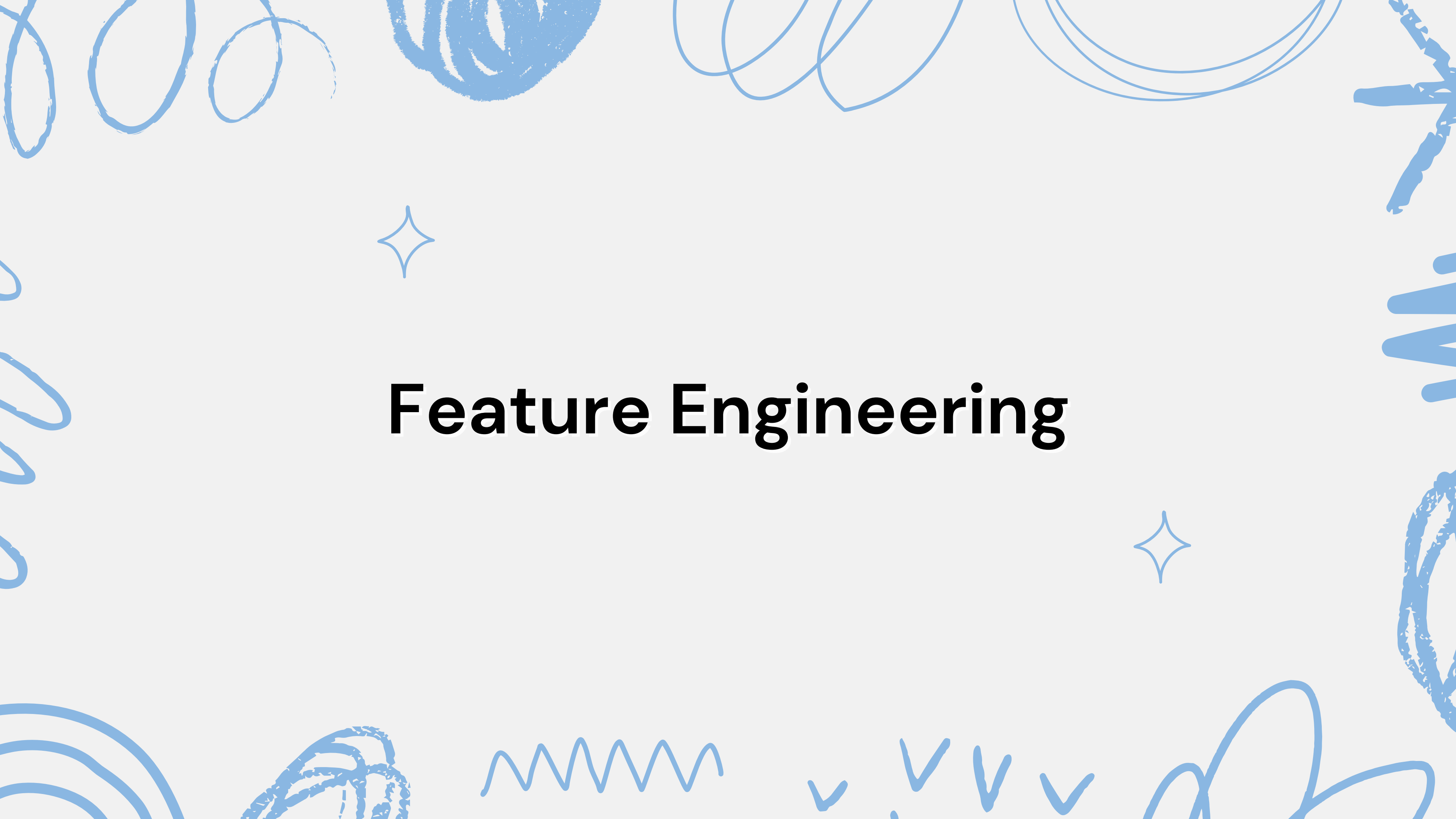
Dimensionality Reduction Accuracy Metrics

Explained Variance Ratio

How much information is preserved

Used in PCA / SVD

Higher cumulative variance = better



Feature Engineering



Feature Engineering

Feature Engineering is the process of selecting, creating or modifying features like input variables or data to help machine learning models learn patterns more effectively. It involves transforming raw data into meaningful inputs that improve model accuracy and performance.

01. Feature Creation

02. Feature Transformtion

03. Feature Extraction

04. Feature Selection

05. Feature Scaling

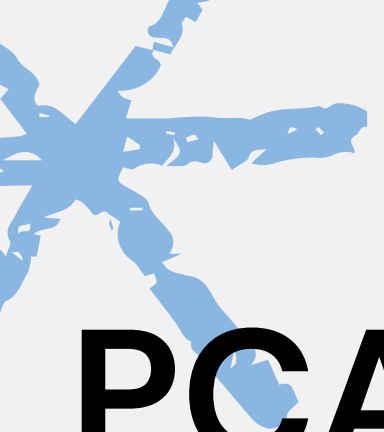




O3. Feature Extraction

Feature extraction is the process of transforming raw data into a simplified and informative set of features or attributes. This reduces data complexity and highlights the most relevant information making it easier for machine learning models to analyze and learn from the data efficiently. It plays an important role in improving model accuracy and reducing computational costs by focusing on essential aspects of the data.



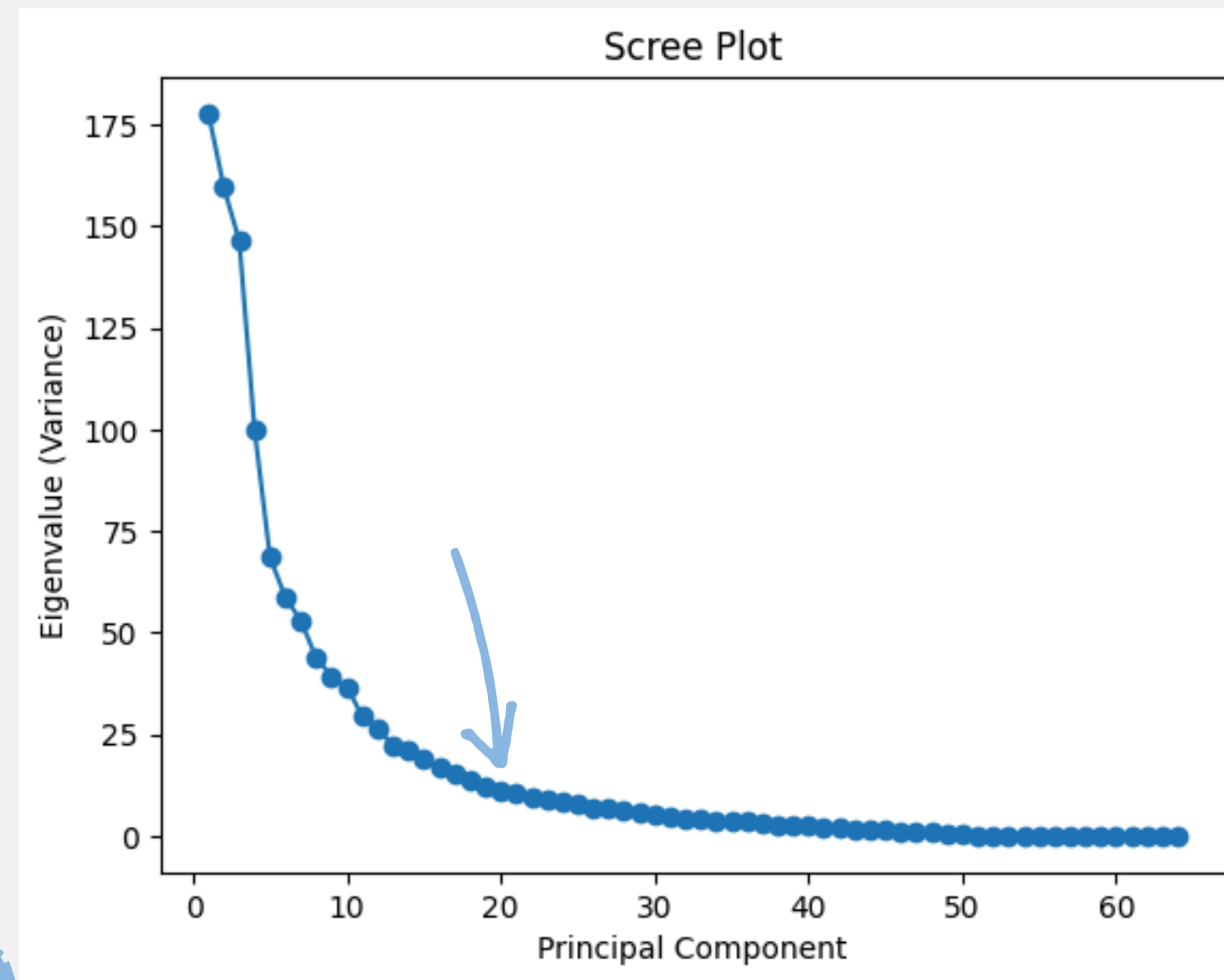


PCA Algorithm Steps

1. Calculate mean vector
 2. Subtract mean vector from each data point
 3. calculate covariance matrix $\frac{1}{n} \times x x^T$
 4. Get Eigen values
 5. Get Eigen vectors for biggest Eigen Values
 6. Normalize Eigen Vectors (optional)
 7. Construct a matrix from the Eigen vectors then multiply it by the matrix of data points
- 
- 

Scree Plot

Helps Choose the best number of components



O4. Feature Selection

Choosing a subset of the original features without changing them.

Correlation

Correlation is a feature selection method that measures the strength of the relationship between each feature and the target variable, usually using a value between -1 and 1. Features with high absolute correlation are considered important because they strongly influence the target, while those with low correlation are often removed. It works best with numerical data and helps quickly identify the most relevant features.

Chi-Square

evaluates whether a feature and the target variable are independent or dependent by comparing observed and expected frequencies. If a feature is strongly dependent on the target, it is considered important and kept; otherwise, it can be discarded. This method is mainly used for categorical data and is especially popular in classification tasks like text analysis.

O4. Feature Selection Cont'd

Choosing a subset of the original features without changing them.

Lasso

Lasso is an embedded feature selection method that adds a penalty to a model during training, forcing some feature coefficients to become exactly zero. Features with zero coefficients are effectively removed, while the remaining ones are kept as important predictors. This allows Lasso to perform feature selection and model training at the same time, making it powerful for handling high-dimensional data.

O5. Feature Scaling

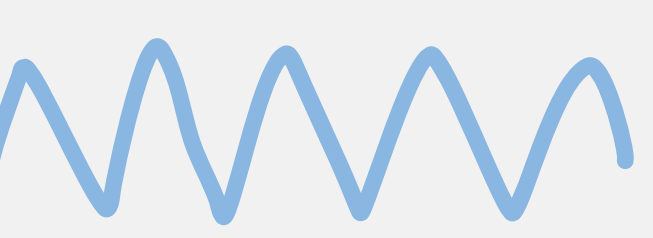
Scaling ensures that all features contribute equally to the model.

Type	Method Description	Sensitivity to Outliers	Typical Use Cases
Absolute Maximum Scaling	Divides values by max absolute value in each feature	High	Sparse data, simple scaling
Min-Max Scaling (Normalization)	Scales features to by min-max normalization	High	Neural networks, bounded input features
Normalization (Vector Norm)	Scales each sample vector to unit length (norm = 1)	Not applicable (per row)	Direction-based similarity, text classification
Standardization (Z-Score)	Centers features to mean 0 and scales to unit variance	Moderate	Most ML algorithms, assumes approx. normal data
Robust Scaling	Centers on median and scales using IQR	Low	Data with outliers, skewed distributions



1. Absolute Maximum Scaling

Absolute Maximum Scaling rescales each feature by dividing all values by the maximum absolute value of that feature. This ensures the feature values fall within the range of -1 to 1. While simple and useful in some contexts, it is highly sensitive to outliers which can skew the max absolute value and negatively impact scaling quality.

$$X_{\text{scaled}} = \frac{X_i}{\max(|X|)}$$




2. Min-Max Scaling

Min-Max Scaling transforms features by subtracting the minimum value and dividing by the difference between the maximum and minimum values. This method maps feature values to a specified range, commonly 0 to 1, preserving the original distribution shape but is still affected by outliers due to reliance on extreme values.

$$X_{\text{scaled}} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}$$